DISCUSSION

SESSION 5B

G. BROWN, *Session Chairman*

Presentation of Papers

I. D. DMITRIEV—*Paper* 5B/1—summarized the main points of his paper.*

K. FURUKAWA—*Paper 5B/2*

In Japan, sodium technology development started in 1957 at the Hitachi Research Laboratory of Hitachi Limited, and the Japan Atomic Energy Research Institute joined with them between 1960 and 1964. Our programme in the initial stage was not very active due to the lack of supporting projects. However, as a result of the energy shortage in Japan, interest in fast breeder reactors has recently been increasing. Therefore, an intense effort is anticipated to build an experimental fast reactor and a prototype reactor.

There are many difficult problems ahead for our programme, but I am sure that our attendance at this conference will help us in some ways to meet these future problems.

P. T. NETTLEY—*Paper* 5B/3

The concept of a sodium cooled fast reactor led to the requirement for data on the rate of corrosion of possible circuit materials in flowing sodium in non-isothermal systems. It also led to requirements for data upon the effect of fast neutron irradiation on the properties of potential cladding materials. Our paper describes some of the data which we have obtained on these two topics. This afternoon I propose to present additional data in order to amplify some of the points which have been made in the paper.

Let us first consider corrosion of steels in sodium. We have shown that corrosion rates depend very markedly upon the temperature of the sodium and its oxygen concentration. At sodium velocities above 3 m/s the rates of corrosion are substantially independent of velocity.

Figure D1 shows the rates of corrosion of fourteen specimens of four main types of steel. These are 316 stainless steel, 18/8 titanium stabilized, $2\frac{1}{4}\%$ chrome/,molybdenum with small additions of niobium and titanium, and a nickel based alloy containing 40% nickel which we call PE16. Its precise

* It is regretted that it was not possible to report the interpretation of the presentation of this paper owing to interference in the receiver.

composition is given in Table 2 of the paper. After an initial period all the steels exhibit very similar corrosion rates which are linear with time. The corrosion rate is about 5×10^{-3} cm per year in sodium containing 25 ppm of oxygen at 650°C. The effect of oxygen on the corrosion rate is very clearly shown in Fig. D2.

In Fig. 5 of the paper we have shown how the corrosion rate changes when the oxygen concentration is taken from a low level to a high level. In Fig. D2 we show the effect of change in the oxygen concentration from a high level

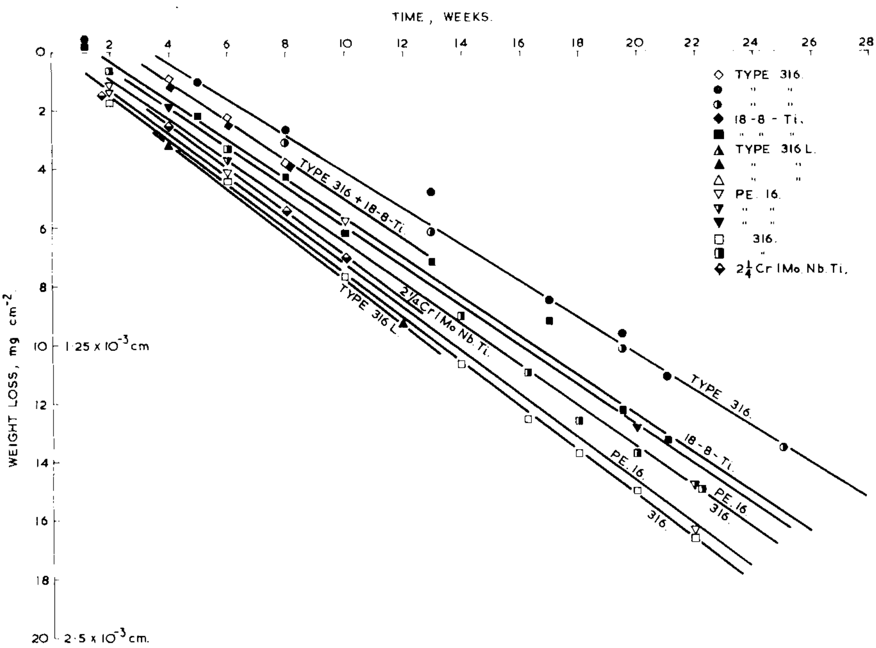

FIG. D1. A comparison of the behaviour of various steels in flowing sodium at 650°C, oxygen concentration about 25 ppm, sodium velocity 5–8 m/sec^{-1}.

to a low level. You will see that the change in oxygen concentration is immediately accompanied by a much lower corrosion rate and the corrosion rate of the specimens is the same as that of new specimens inserted at the time when the oxygen level was lowered. Thus we see that the specimens do not remember the effect of the high oxygen concentration that they saw earlier. The change in corrosion rate is dictated by the instantaneous oxygen concentration. In contrast to the behaviour of stainless steels, nickel alloys, which contain a high concentration of nickel, show corrosion rates which are substantially independent of the oxygen concentration, but are very markedly dependent upon velocity.

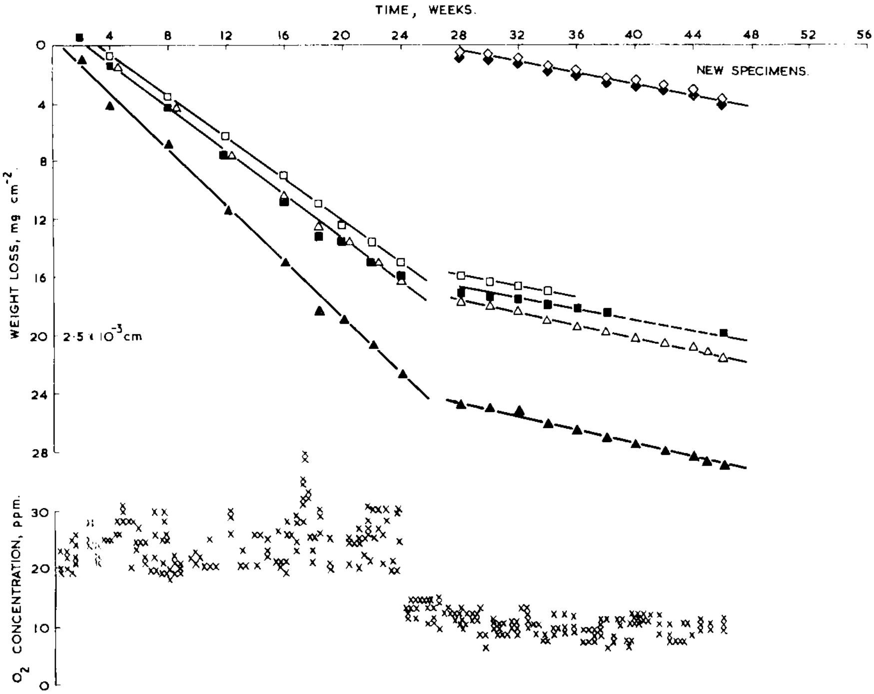

FIG. D2. The effect of reducing the oxygen concentration in the sodium on the rate of metal loss from 316 stainless steel; temperature 650°C, sodium velocity 8 m/sec^{-1}; specimens *in situ* during the change.

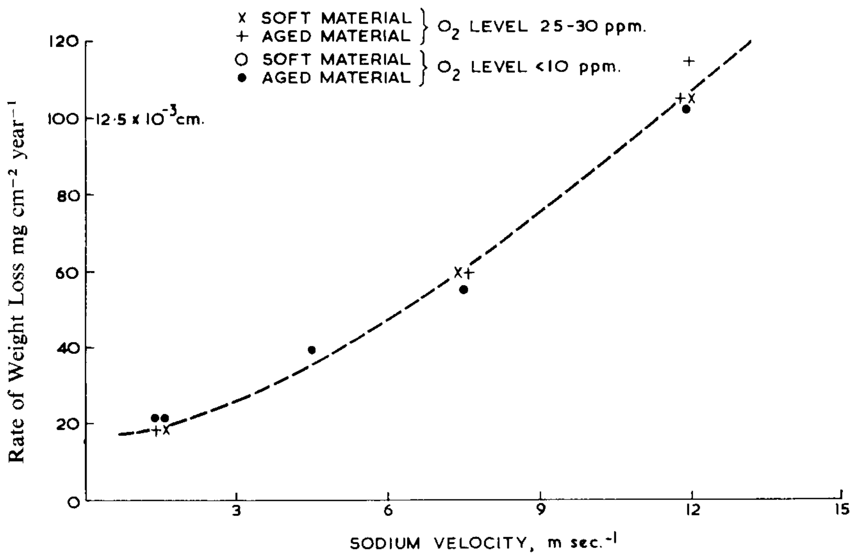

FIG. D3. The effect of sodium velocity on the rate of metal loss from Nimonic 80A at 650°C.

Figure D3 shows the behaviour of an alloy called Nimonic 80A which contains about 80% nickel. The material (see Table 2 in Paper 5B/2) exposed to sodium at 650°C contained one lot of specimens at 10 ppm of oxygen and others at 25 ppm of oxygen, and the results show that oxygen has very little effect, but that sodium velocity dictates the rate of corrosion. At

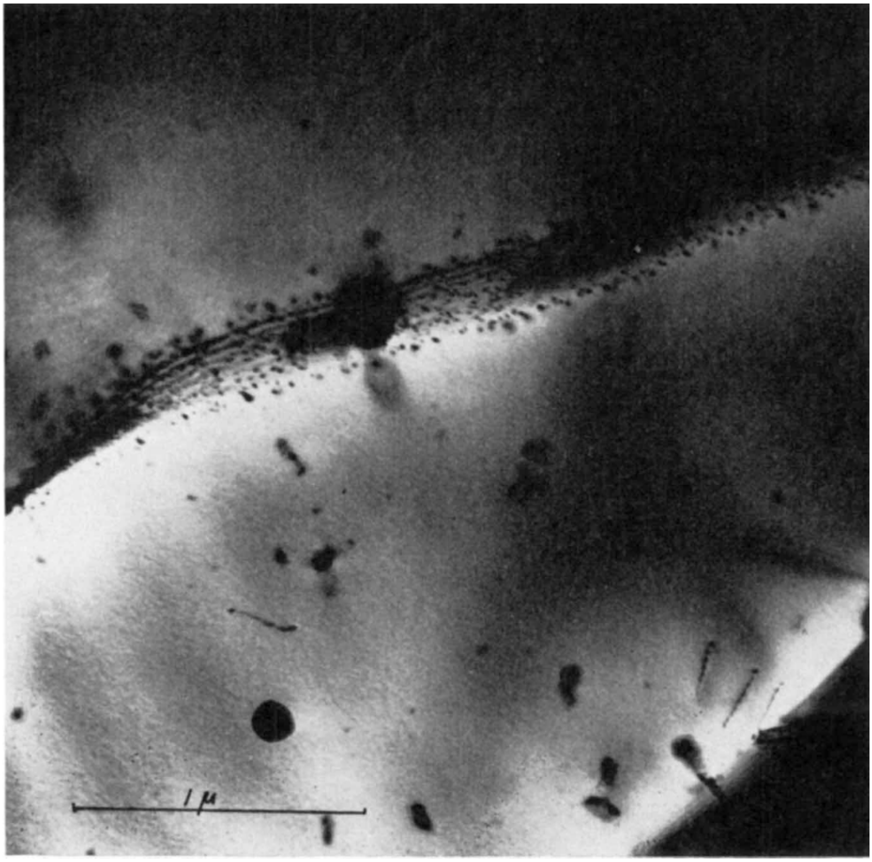

FIG. D4. Grain boundary NbC phase in FV548 stainless steel.

high velocities of about 12 m/s, the corrosion rate is rather alarming, being $12\frac{1}{2} \times 10^{-3}$ cm per year. The graph also shows that the metallurgical condition of the material is not important. You get the same corrosion rate with solution treated material and with solution treated and aged material.

Turning to high temperature embrittlement of cladding materials, in the paper we have stated that from thermal reactor irradiations we believe that a steel called FV 548, which is similar to 316 material but which contains 1%

niobium, is more resistant to irradiation induced by high temperature embrittlement than 316 material. We have stated that we believe this is due to the presence of niobium carbide which strengthens the grain

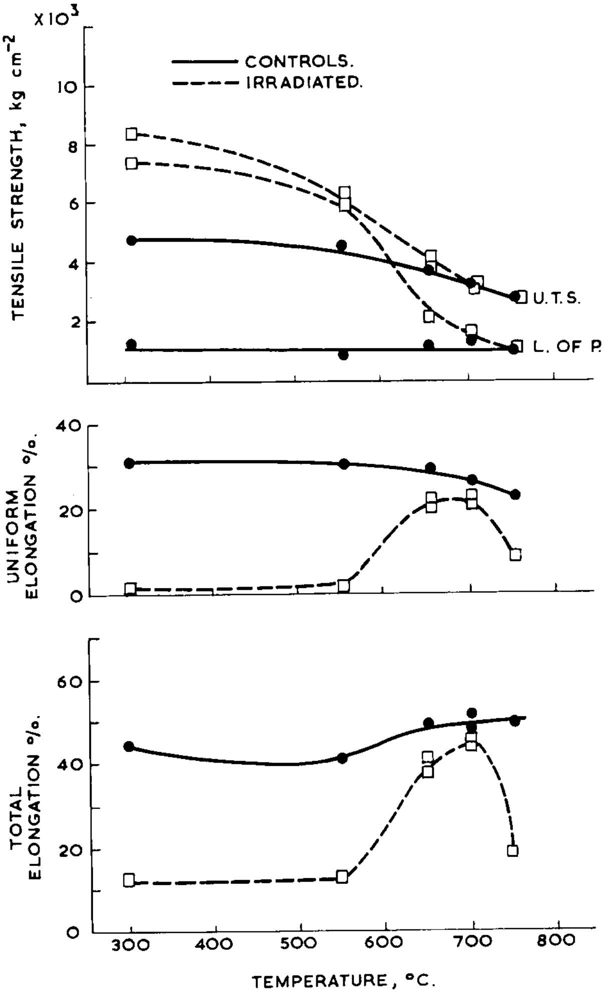

FIG. D5. The tensile properties of solution treated and aged FV548 stainless steel after irradiation to a total dose of $2{\cdot}2\times10^{22}$n/cm^2 at 300°C in DFR (initial strain rate $5{\cdot}8\times10^{-4}$ sec^{-1}).

boundaries, and this niobium carbide comes down at the grain boundary when the material is aged.

Figure D4 shows an electron-microscope photograph of a grain boundary of FV 548, and shows the niobium carbide precipitate at the grain boundary. We have irradiated FV 548 material in the Dounreay fast reactor at temperatures of about 300°C and Fig. D5 shows the short term tensile properties of the material after it has been irradiated.

You will see at the top of Fig. D5 that point defect damage gives a marked increase in the U.T.S. and the limit of proportionality at low temperature.

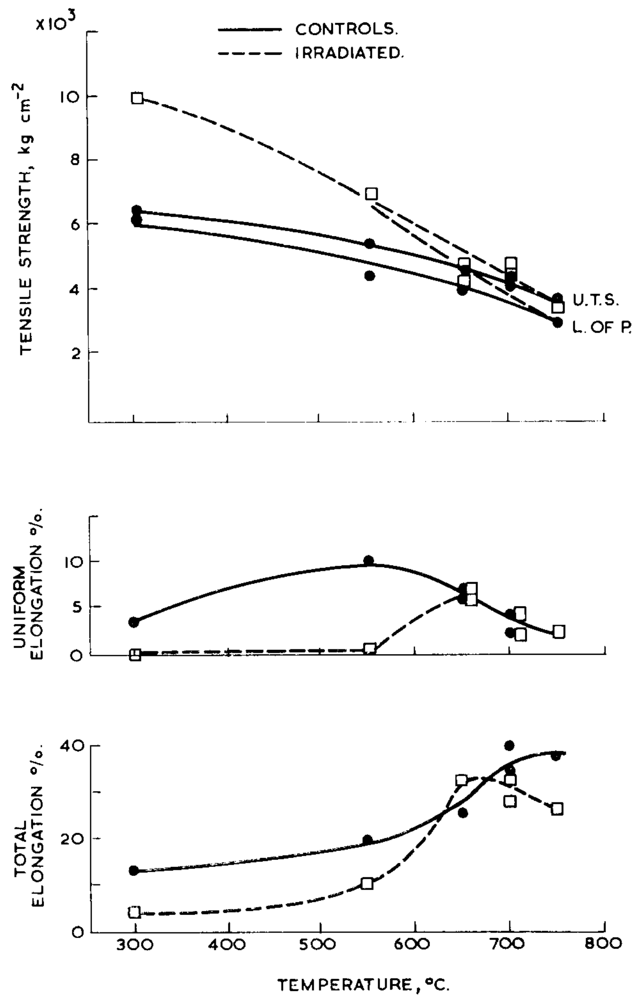

FIG. D6. The tensile properties of 20% pre-strained FV548 stainless steel after irradiation to a total dose of $2{\cdot}2 \times 10^{-22}$ n/cm at 300°C in DFR (initial strain rate $5{\cdot}8 \times 10^{-4}$ sec^{-1}).

This decreases as the temperature increases. Similarly the point pefect damage gives a marked drop in uniform elongation up to 500° C. High temperature embrittlement is not particularly serious until you go to temperatures of 750°C. This material was irradiated to a total dose of $2{\cdot}2 \times 10^{-22}$ n/cm^2 and the strain rate was $5{\cdot}8 \times 10^{-4}$/s. Its good behaviour is not entirely due to niobium carbide at the grain boundaries, because we have also irradiated material in the cold worked condition, and it was not aged, so it did not contain niobium carbide at the grain boundaries.

Figure D6 shows the results after the irradiation of this material, and if anything its resistance to embrittlement is even better than solution treated and aged material. We get similar behaviour at low temperatures, point defect damage raising the U.T.S. and limit of proportionality, but this is annealed out at temperatures of 600°C and above. We see again that point defect damage gives a marked reduction in ductility at low temperatures, but this damage is annealed out at 600°C and above. Even at temperatures of 750°C the high temperature embrittlement is quite small.

To sum up, we feel at the present time that providing low oxygen concentrations can be maintained in the sodium, no problem should arise due to corrosion of steels in a fast reactor circuit at temperatures up to 650°C.

If reasonable precautions are taken to avoid putting too much carbon in with the initial sodium, and care is taken in choosing materials which do not decarburize in sodium, there should be no serious problem due to transfer of carbon. Good progress has been made on the effects of neutron radiation on creep, yield, and ductility of potential cladding materials, but it is obvious that much more work is required before we can make a final choice.

J. J. Morabito—*Paper* 5B/4

(Before presenting his paper, Mr. Morabito paid tribute to H. W. Savage, his colleague and co-author, who died in December 1965.)

Development of sodium components in the United States has been conducted primarily by the laboratories and industrial organizations involved in the design, fabrication, construction, or operation of the major United States sodium cooled reactor plants. As a result of design and operating experiences with these plants, sodium components are being both proved and developed, and requirements for additional component development efforts and test facilities are being identified.

This paper discusses the experience and current status of the Sodium Components Development Programme and some of the significant results of the evaluation of the operation of Hallam and SRE components since the ANS Topical Meeting on Fast Reactor Technology in April 1965. Current operating experience at EBR-II and Fermi is included in other papers presented at this conference.

K. G. Eickhoff—*Paper* 5B/5

The programme that I described in the paper was undertaken to provide the basic technology necessary for the design and operation of sodium cooled circuits in a prototype fast reactor. These coolant circuits will operate at temperatures of up to 650°C and velocities of up to 5 m/s.

In addition to establishing this basic technology, our engineering development programme aims to provide information for the design of individual reactor components. Development work has covered mechanical sodium pumps, fuel elements, flow distribution, gas entrainment, and sodium/water reactions.

As a result of our experimental work, which is described in the paper, we are confident of the possibility of operating sodium circuits under conditions which we envisage in our prototype fast reactor. In addition, our programme for the development of reactor components is well advanced and no engineering problems have been uncovered which would in our judgement prevent the realization of our prototype fast reactor design concept.

M. ROBIN—*Paper* 5B/6

Mr. Robin summarized the main points of his paper.*

Discussion

S. SIEGEL (Atomics International): To what extent are the peak pressures which have been observed in the sodium/water reaction experiments truly associated with the reaction between sodium and water? To what extent are they really something like the water hammer phenomenon when high pressure water first enters the sodium system?

D. G. H. LATZKO (The Netherlands): Referring to Fig. 6, Paper 5B/6, it shows that the actual pressures measured at the location of the sodium water leak were greater by a factor of about two than the theoretically calculated value. Have the French been able to adjust their theoretical model in such a way as to get better correlation between calculation and experiment?

M. ROBIN—*Paper* 5B/6: We are not able to calculate the peak pressure. In fact we have no model which is satisfactory so far, but we are trying to get one.

J. ALLEN—*Paper* 5B/5: I am afraid that I can add very little to what has been said on the question of peak pressure. I do not know the answer to this. We have recorded, as have most workers, high pressure immediately at the breakage or simulated failure of the tube. This could have been due to the rapid evolution of hydrogen. My opinion is that it may be associated more with the water hammer than with sodium/water reaction effects.

J. TATTERSALL (TNPG): I should like to know whether anybody has evidence of the importance of the various factors which take place in the reaction following a small leak on the propagation to neighbouring tubes. Is the

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temperature of the neighbouring tubes important and, if so, is the rig adequately simulating the cooling which will take place in practice with water flowing on the inside?

J. J. MORABITO—*Paper* 5B/4: I think that the immediate answer to the question is no. That is why we have an experimental programme to separate these variables, and find out which are important. The experiment I referred to had two different kinds of heat transfer conditions prevailing. Temperature at the reaction point was about 100°C different, and one was on the water cooled tube and the other was on the sodium cooled wall. This may mean something in terms of the rate; I think it does. That is why we look at temperature as one of the variables.

J. ALLEN: On the question of propagation due to temperature, on one of our more recent reactions we have noticed failure of a tube in the vicinity of the reaction, and we believe this is not due to the corrosive effect caused by jets of steam, but solely due to the increase in local temperature of the tube which has raised the pressure inside the tube. In our last experiment we met pressures of 4000–5000 lb/in^2 and this in itself, associated with the local high temperatures, appears to have caused failure of an adjacent tube. This is a different mechanism from those which have been reported in other papers where we have a jet of water/steam which causes corrosion, and which reduces the wall thickness. So we have this second mechanism.

H. HARDE (Germany): For 2 years we have been conducting a series of experiments related to water/sodium interactions, both for large and small leakages. As far as the latter are concerned, the experiments are of a preliminary nature and they tend to indicate the same results as Mr. Morabito has outlined, in that the distance between the small hole and the plate on which the jet impinges is very important from the point of view of the time required to achieve substantial wash out. Secondly, the size of the core is important. One causay that there is no problem in penetrating a plate of steel in 60 seconds through a thickness of several millimetres if we choose appropriate parameters.

With regard to the large leakage experiments, we are interested at present in the design and construction programme for a 20 MW experimental reactor, particularly of the steam generator design which, in many respects, is similar to what the French are pursuing, namely a hairpin-shaped tube where there is one wall separating the water and the sodium. One can call this a one-dimensional steam generator, and we have directed our attention mainly to this type of geometry for the time being. We have conducted up to sixty or seventy experiments which involve about 6–10 kg of sodium. Since the ratio of cross-section of the water tube as compared to the diameter of the sodium in the pipe was large, the initial pressures registered by the quartz transmitters were very high. We have recorded pressure surges lasting as long and as high as 1800 atm. After 50–200 msec these pressures have been at steam temperature.

We have endeavoured to describe the results we obtained by theoretical means, and we were successful in identifying the main contributory factors which govern the reaction. Our conclusions are that for the first 3 msec the reaction is carried on by the direct contraction or diffusion of reacting particles through a rather small layer and we could theoretically duplicate the experimental results within the first 2 msec. Our theoretical interpretation of the first pulse tends to indicate that it is really a reaction pulse. After 2–3 msec it becomes obvious that the reaction is being carried on only by reaction taking place between the sodium left on the walls and the water that is intruding. As a matter of fact it can be duplicated in such a way that in each case the layer thickness derived from theory is between 0·1 and 0·2 mm. In that way it can be shown that a water reaction, except for shock wave reflection, can be reproduced theoretically.

Finally, I might mention an interesting result which might play an important role with regard to maximum pressure. If the sodium temperature is somewhat higher than the water that has been injected, and if the water is at saturation, after a certain time secondary reaction is observed which can be traced back to a sudden release of heat that is in the walls adjacent to the sodium. There is a steam blanket at the surface of the walls when water protrudes into the tube. After a while, when disturbance in the system occurs, this steam blanket is destroyed, eliminating the thermal insulation, and in a short duration the whole latent heat in the walls is added to the system giving rise to steam flashing. This might give a pressure rise as high as in the initial reaction.

S. SIEGEL: Have any of the experimenters tried a blank experiment with water and water to look for the magnitude of this initial pressure pulse? I must confess that Dr. Harde's description seems to be a very detailed explanation of what has been observed, but I am curious to know whether there has been experimental examination of the question by using two non-reactive fluids.

J. J. MORABITO: One might ask what is the significance of the question as to whether it is caused by the initial rapid release of the water or the water reaction. The end product, as far as the equipment is concerned, might be the same. I am curious why Dr. Siegel asks the question.

S. SIEGEL: There has been a great deal of psychological concern about sodium/water reactions. No one has expressed concern about high pressure system relief into low pressure system. You normally take care of this by rupture disks, which has been the approach made on sodium/water reaction rigs. Rupture disks seem able to take care of the reaction, and I thought that if we could demonstrate by experiment that we were looking at hydraulic effects rather than unpleasant chemical reaction effects, we might all be further reassured.

K. G. EICKHOFF—*Paper* 5B/5: I have two questions. One concerns a point in Paper 5B/4. It reports free wastage of stainless steel on the outside of a vessel caused by sodium leak from the vessel but does not say what the temperature is. I would inquire whether the French or Russians have experience of sodium leakage from high temperature vessels.

J. J. MORABITO: I am not sure that it will mean much, because the question might be what was the result of temperature locally, and we did not have a thermocouple on so we do not know what was happening while leakage was occurring. I think that the mechanism is one of sodium hydroxide corrosion. I believe the temperature was 600°F. It was not a high temperature tank, but there may have been high temperature reactions going on locally, and they may have existed for a long period of time.

M. ROBIN: As far as sodium leakage is concerned we had a small leak in a primary pump tank like a pinhole, but after less than 1 hour the outside of the wall was fully corroded, so it is very important to have something like inert atmosphere around it.

A. N. CHARCHAROS (U.K.): Most of the sodium/water reaction experiments appear to have been carried out in static or semi-static pools of sodium. My concern is what is the effect of flowing sodium on leakages, that is, small and large, and the transference of corrosion products to other parts of the secondary sodium circuits?

J. J. MORABITO: You note in Paper 5B/4 the list of experiments I mention were under flowing sodium at velocities of 1·3 ft/s. I do not think that I would attempt to arrive at any conclusion between these and static experiments, because the conclusion would be that it is worse locally with flowing sodium, and I do not think this makes sense logically. I think it is within the experimental error.

J. ALLEN: We did an analysis of the effect of flowing sodium in relation to our own experiments which have in the main been large leaks. We came to the conclusion that flowing sodium or water on the inside of the tube would have little effect on reaction in the case of a large leak, by which I mean a complete tube fracture. I think it is evident from the results we have had that sodium/water reaction is accompanied by the evolution of hydrogen, and when you get large reactions you virtually isolate the flow of sodium locally. Therefore, I feel that sodium flow has little effect on large leaks. On small leaks sodium flow undoubtedly will have an effect, and there it is a question of degree. If the leak is such that in normal circumstances it might act as a jet on an adjacent tube, say, 10 ft/s might well prevent that particular jet from causing damage to an adjacent tube. If, of course, you make the jet bigger, the sodium flow would have less effect.

THE CHAIRMAN: This leads to another question which has been raised in the papers, namely, the difference between the rates of corrosion at different velocities. There is some evidence for an increase in corrosion given by Dr. Nettley in this extra information, and I gather Mr. Morabito corrected what he put in his paper by saying he had further information of an increase in corrosion with velocity.

J. J. MORABITO: The point I was making was that I believe Dr. Nettley said, 'above a certain velocity'. I would say that above 9 ft/s there was no effect. Our experiments were conducted at 7 and 21 ft/s respectively, and we found that the effect was essentially linear with velocity. But the question is, is velocity or Reynolds number the important parameter. I think there is some question in this area.

P. T. NETTLEY—*Paper* 5B/3: I agree with what Mr. Morabito said. We find a velocity dependence in our geometry at velocities below 3 m/s but above that velocity we find no velocity dependence. It may well be that the proper parameter to use is Reynolds number, so that if we used a different geometry we would find a different cut-off point. You can only compare the results from two laboratories if they used identical geometry, or if a proper relationship between corrosion and Reynolds number is established.

J. F. W. BISHOP (UKAEA): It would be interesting to have some discussion on the point whether it is sodium velocity or Reynolds number which is the important parameter in corrosion. Do any of the corrosion experts have information on the point?

C. TYZACK—*Paper* 5B/3: It is dangerous to make statements about corrosion in our present state of knowledge. I do not think the position is very clearly defined at the moment with regard to iron and steel, but there seems to be some indication that if you plot mass transfer rate against Reynolds number to the point eight, there is a levelling off and the height of the plateau is a function of the oxygen level.

The activation energy for the reaction which we have measured at high velocities, which is about 17 kcal, suggests it cannot be principally a diffusion controlled reaction, because if it were, you could account for roughly 4000 calories on the diffusion dependence on temperature, and about the same amount on the heat of solution of iron and sodium which would give us about 8000 calories. So we would still have about 9000 calories to account for. I think that this figure of 17,000 which we have measured probably represents a surface reaction, and we would expect this value to be sensitive to the actual oxygen level in the sodium; but we do not have good enough numbers to differentiate different slopes.

I think it has been known for some 10 years that there are something like two or three orders of magnitude of discrepancy between the calculated rates, taking known diffusion coefficients, and observed rates. I think we are coming round to the view that this is primarily because the sodium is nearly saturated with respect to iron.

I think you can account for a large part of this if you assume that sodium is 99% saturated with respect to iron in the circuit. A further pointer in this direction is the effect you find downstream—an effect which has been found by GE workers and confirmed by us—which suggests that you may not be a great distance from saturation. This behaviour contrasts quite strongly with that of nickel alloys. There is a big difference, because if you take a pure nickel specimen and a stainless steel system, there is a big activity difference between pure nickel or sodium adjacent to the pure nickel and that in the sodium in the rest of the circuit which will be controlled roughly at that of the nickel activity in stainless steel. So we find more than a linear dependence on Reynolds number here.

The other interesting point in relation to nickel is that it does not appear to be sensitive to oxygen level in the sodium, whereas iron does. I think it has been shown that it is not possible to form nickel–sodium–oxygen compounds in the presence of sodium, whereas if you have a high enough oxygen level with iron, you can form a discrete compound. That is the essential difference between the two systems.

K. FURUKAWA—*Paper* 5B/2: Dr. Nettley said that the corrosion phenomena of steels mainly depend on temperature and oxygen concentration. I think that the oxygen concentration will be assumed from the plugging temperature which, in some cases, means hydride concentration. I think that we should refer to the activity of oxygen rather than oxygen concentration. I am interested in the effects of other impurities, which certainly exist, metallic and nonmetallic impurities in the same order of concentration. One can expect them to have an influence on oxygen activity at least. Has Dr. Nettley any evidence of this? There is, for example, hydrogen, carbon or calcium. Is the corrosion data dependent on the raw metal of sodium or not?

I am now attempting to do some corrosion work using more controlled purity sodium in order to obtain more reproducible data.

P. T. NETTLEY: The impression that oxygen content was determined from the plugging temperature is incorrect. Oxygen content was measured by chemical analysis. The question was do we know of any effect of other impurities in the sodium on the rate of corrosion, and I will ask Dr. Tyzack to reply.

C. TYZACK: I cannot say anything in relation to stainless steel, but as far as refractory metals go, some work done by Thorley suggested that under conditions of high carbon you get quite an effect on weight changes, and X-ray examinations show the presence of niobium carbide. We had pure niobium, we had niobium carbide and nitride as well as complex sodium niobium oxides. I think that under these conditions one would certainly expect effects due to other impurities besides the oxygen.

If I can throw the question back to Dr. Furukawa, he said that in his work he saw not only enhanced loss of chromium with stress, but enhanced carburization. Would he not expect, if he had enhanced carburization, to slow down the loss of chromium by the formation of chromium carbide?

K. FURUKAWA: I think there is some confusion. You refer to our preliminary work, and in that case we used dirty sodium. I think that we should now carry out an examination of that programme.

C. TYZACK: Did you have evidence of oxide films on steel with dirty sodium?

K. FURUKAWA: I think so.

C. TYZACK: It is presumably a sodium chromite film. The other thing about your work is that you are getting mass transfer rates about five times higher than we observed in our loops. The only obvious possible explanation is that you have a much bigger volume of sodium in relation to surface area of the pipe work. Consequently, it may be that your sodium is relatively unsaturated with iron. It has less iron in it, and therefore one has a bigger driving force for mass transfer. Have you any analytical figures for iron in sodium?

K. FURUKAWA: Not quantitatively, no.

A. G. FRAME (UKAEA): Can we with confidence use the results that are being produced in the design of reactor circuits? We have had indicated that results might be very sensitive to impurity level or element level in the sodium and, in all reactions, rarely is the volume of sodium truly reflective of what there is in a reactor. If the volume of sodium does not truly reflect what one has in a reactor, are results with a reactor likely to be worse or better?

P. T. NETTLEY: I think that if the designer can tell us what the oxygen level in his plant will be, we can tell him that he can use our results with confidence, because the only way in which we can see they will be wrong is that they will be pessimistic. The reason is that, if you get saturation effects, you will get much lower corrosion rates than the values we are getting for short lengths in our apparatus The types of corrosion rate we get in small loops are very similar to the values we get in the large loops which Dr. Eickhoff described. So you can use the data, and if they are wrong they will be pessimistically wrong.

J. ALLEN: We have made some assessment of corrosion quantities from the point of view of reactor design. If, as Dr. Nettley says, we use his rates on a *pro rata* basis, can we equally apply those rates to low temperature parts of the reactor system? Can we say that our heat exchangers at one end are 400°C and the corrosion rate in that region is of the order of one-fifth of 0·001 inch per year under these conditions? Do we add up total corrosion and say it is half a ton a year?

C. TYZACK: What you are measuring in our type of loop or the GE loop must be the slowest reaction in the system; because we suspect the iron is getting near saturation, the slowest step must be the deposition step. We know very little about this process, and it is a very difficult one to study, but if the deposition of reaction was extremely efficient, you would keep the iron level in sodium very low. But as it rises near saturation, it must indicate that the control step is the deposition step.

A. E. GOLDMAN: I believe it is very important to talk about oxygen activity in sodium rather than about oxygen content. When Dr. Tyzack speaks about the saturation of iron in the sodium, he is concerned with iron or possibly iron oxide, or some complex oxide saturated in the sodium. Under those conditions one has a fairly good idea of what happens, but one does not have an idea of what happens when oxygen exists in many other forms in the sodium. There can be sodium hydroxide or sodium carbonate in solution, and oxygen in the sodium in these different forms has different activities and therefore will cause corrosion rates which are quite different. Therefore, as long as one knows what the oxygen content is, Dr. Tyzack must qualify his statement that it is not total oxygen effects in the sodium, but oxygen activity in the sodium. With zirconium particles floating around, the oxygen is so tied up with the zirconium that it is incapable of further corroding iron. On the other hand, if you looked at the total oxygen you would recognize it as zirconium oxide.

M. J. R. LEDUC—*Paper* 5B6: Modern generators have been tested on a small scale loop. Is so small a loop adequate for the testing, or is it intended to build a larger loop with larger characteristics for the future?

A. I. LEIPUNSKII—*Paper* 5B/1: The tubes are of natural size. Only the number of tubes is small. All the parameters are the same, so we shall adequately know the thermal properties of the steam generators. There is also the important problem of vibration, but we hope to solve this in another experiment. We shall build some new facilities, but I do not know when. They will be for testing some new designs of steam generators.

H. HARDE: We have operated sodium heat transfer systems which duplicate the complete reactor system. They can transfer 5 MW(th) to a steam generator.

We believe that one of the main problems associated with sodium heat generators can be solved by operating small units. The technological differences between flue gas heated or steam generators and sodium heated generators are that in the latter you can generate heat flux superior to that in gas heated generators; and, although the overall performance of steam generators is successful, in monitoring the steam for hydrogen we observed an amount of hydrogen in the steam which was indicative of continuous magnetite formation, which may pose problems in so far as it may be due to local corrosion.

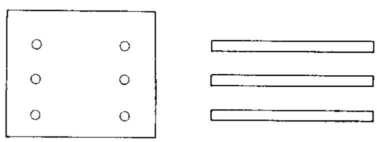

FIG. D7

W. B. HALL (UKAEA): My comments concern the question of heat transfer with liquid metals in clusters of rods. I have a specific comment on Paper 5B/5 in which it is said that the ratio of two eddy diffusivities is measured, one being for mass in air and the other for heat in sodium. I should like the authors to confirm that the quantity ε_M, which is a unique function of Reynolds number, is the quantity diffusivity over kinematic viscosity and not diffusivity alone.

The other comment concerns the diffusion of heat across a cluster of fuel elements cooled by liquid metal. If we take as an example a cluster of rods (Fig. D7) and if we look at the rate of temperaure penetration, if sodium takes T to flow through the element, the temperature penetration is the square root of diffusivity $\times$ T. If the time in a fast reactor is 0·1 second, and I estimate diffusivity to be 1 cm^2/s, then the distance is of the order of 0·3 cm, which means that the temperature effects are not diffused across the sodium. I wonder therefore whether it will be necessary to introduce some better mixing by a secondary flow. I think that the matter is worse with liquid metals than with gases, because the effect with a gas is completely swamped by a large temperature drop.

K. G. EICKHOFF: Our experiments to measure diffusivity of mass in air and heat in sodium have been condensed in our description. I believe that what Professor Hall suggests is substantially correct; but we shall give a more detailed analysis in thc published proceedings* which will clear up the confusion which seems to have been caused by our limited account in the paper.

As to the comment that we need more mixing of fluid in various channels in the fuel element than is given by diffusivity of heat, it is true, and it is possible to enhance the mixing between one channel and another by the precise shaping of the grid which is there primarily to hold up the fuel element, but which can be made to fulfil the secondary role. We are investigating the amount of mixing which can be produced by the variation of the precise shape of these fuel elements.

J. WENGLER (W. Germany): What happens if the fuel flows stops, with fuel which gives heat to the sodium?

A. G. FRAME: In the prototype fast reactor the pumps have sufficient inertia in the system so that the heat produced in the fuel element can be removed in the normal heat exchange system as the power of the reactor falls off. When the reactor shuts down, heat is removed by gravitational inertia in the pump system.

K. WIRTZ (W. Germany): Dr. Wengler believes that the sodium will be superheated to a considerable degree. The superheating of sodium is possible, and the considerable heating of sodium would mean a high rate of release of heat, and it could mean the rapid expulsion of the sodium. Has anybody given thought to that problem?

A. G. FRAME: I think you will find in Paper 3/3 that we have considered this problem, and it is currently our major concern on subassembly safety.

G. DENIELOU (CEA): What is the basis of the trust which Dr. Nettley seems to have regarding the behaviour of stainless steel cans with or without the

* See K. G. Eickhoff's written reply to Professor Hall (p. 936).

addition of niobium in the integrated fast flux range between 3×10^{22} and 5×10^{23}?

Secondly, what is the integrated fast flux maximum which gives rise to the observations on the matter of cans in the United Kingdom?

P. T. NETTLEY: There are figures in Paper 5B/3 which show that the maximum dose we have been to is 6×10^{22} in the DFR spectrum. You will also see that we have plotted the parameters for stainless steel in Fig. 7, Paper 5B/3, and the ductility is decreasing linearly with the log of the total dose. Therefore, unless some new phenomena appear, it seems as though the difference between behaviour at 5×10^{22} and 5×10^{23} will not be large if we extrapolate the present rates linearly. The DFR has a harder spectrum, and 5×10^{22} in DFR in terms of helium production is something like twice that value in a very large reactor. In other words, 5×10^{22} could well be equivalent to 1×10^{23} in a large fast reactor.

H. STEHLE: In Paper 5B/3 the tensile elongation illustrated in Fig. 7 shows that, for irradiated stainless steel at high temperature, tensile ductility is not so low. Could he explain why, in spite of this, the fracture of pins with much less plasticity has been found?

P. T. NETTLEY: It is well known that the ductility of a tube is far less than the ductility of a tensile specimen. It is an extension of the same phenomenon —you cannot get into the tertiary creep range in a tube. You only get uniform elongation and very little of the non-uniform elongation that you get in a tensile specimen.

H. STEHLE: How can you relate your experiments and results to the layout of a fuel pin?

P. T. NETTLEY: I do not think we have attempted to do that. We are attempting a broad programme on cladding materials to find out the effects of fast neutrons on the properties; the object of the programme is to be able to understand what happens to a cladding in a fuel pin. The data are not sufficiently comprehensive to give all the answers at this particular point of time. All we have given is the data we have, and you will have to wait a little longer before we can answer your questions.

J. F. W. BISHOP: In the evaluation of this reactor fuel element we have found a better tie-up between the overall cladding data and the fuel performance than I personally have met before in fuel element development. The normal situation in fuel element development is that you have a fairly large factor between materials data and fuel element performance. In the results which the Russian authors have quoted in Paper B5/1 and in the results quoted by Dr. Lawton yesterday (Paper 4B/4) failures in fuel pins have been spoken about at 2% strain, and this is the sort of number which we would expect to find from our overall cladding programme.

H. STEHLE: How do you account for the vast dose?

P. T. NETTLEY: In DFR irradiation we count all neutrons.

H. STEHLE: What is known about the influence of irradiation on the rate of weight loss in sodium?

P. T. NETTLEY: The answer to that is not very much. It is difficult to think of a mechanism whereby irradiation should affect the rate of corrosion. Corrosion rates in the Dounreay fast reactor (DFR) are low because velocities are low. The activity of the oxygen in sodium is also low, and it is difficult to get a correlation between DFR and laboratory work. But the corrosion of stainless steel in the flux of the DFR is very low indeed.

J. P. ANDRE: Dr. Nettley states in Paper 5B/3 that increasing stress linearly with time does not provide repeated increments of primary creep. Does this mean that actual creep tests under linearly varying stress have given the same kind of trends as the strength calculated using the appropriate secondary creep rate level?

P. T. NETTLEY: The answer is yes, but I should emphasize that the experimental work has only been performed on one material, namely 316 L type material with low carbon.

N. G. WORLEY: I should like to turn the discussion towards the boilers rather than the fuel elements. We have heard a great deal today about sodium/water reaction experiments which are being carried out in various parts of the world, and from the design point of view these experiments are very important, and the data they are providing used to design vessels and tubes that we employ. But what we should like to know more about from the design point of view of boilers for fast reactors—and it is true to say that the success of a reactor may well depend on obtaining some of this information—are the circumstances which can lead to tube leaks. We know that some of these are complicated, such as vibration, stress corrosion, thermal cycling and fretting, and so on; but I should like to ask whether the authors are carrying out any experiments to define the conditions under which ferritic austenitic tube joints in boilers are acceptable in sodium.

H. HARDE: Our experience in this is related to two types of experiment. First, we were interested in how far that kind of joint would be permissible in a sodium system at all. One may anticipate some thermal cycling, so we conducted a series of thermal cycling experiments between ferritic and austenitic steel at temperatures between room temperature and 600°C. We went up to 600 cycles without the slightest sign of fatigue failure. In one of those steam tubes we have an inconel ferritic seam weld, and although it is not as demanding on the weld since the expansion of inconel is less, we have conducted a similar experiment and the thermal cycling has not led to any problem so far.

J. CHAMPEIX (CEA): What does Dr. Nettley think about the possible use of cladding material of vanadium alloys or other refractory metals in sodium cooled fast breeders?

C. TYZACK: It is a long time since we did any work on vanadium, but I think it is in much the same category as niobium. You could not use it above a temperature of 450°C with cold trapping but would have to use hot trapping. I am not very impressed with the alloys which have been put forward.

J. CHAMPEIX: We do not wish to use a hot trap although resistance to corrosion in the event of hot traps is good. I wonder whether additional elements in vanadium could improve the resistance to corrosion?

J. T. RATIER (CEA): In Paper 5B/3 Dr. Nettley states in connexion with the nickel alloy PE.16 that "precipitation of the hardening phase during irradiation appears to inhibit the processes which lead to loss of strength and ductility at high temperature." In my opinion this assumption does not appear very clearly in Fig. 9, Paper 5B/3. What precise effect has he seen with this alloy, and what mechanisms are involved in the change in his view?

P. T. NETTLEY: If you look at Fig. 8, Paper 5B/3, you will see that ductility even in the tensile specimen is zero at 700°C. So that if you are to make any demands on ductility, it looks as though you might be in trouble around the 700°C mark. This is a low temperature irradiation. We appear to observe less loss of ductility when we irradiate PE 16 at high temperatures. Results of the high temperature irradiations are not given in the figures.

Written Discussion

W. B. HALL (U.K.): I would like to make a specific comment on the measurements of eddy diffusivity for heat in sodium and mass in air which are described in Paper 5B/5 and a general comment on temperature distributions in clusters cooled by a liquid metal.

The authors state that the ratio of the eddy diffusivity of mass, ε_M, in air to that of heat, ε_H in sodium at a similar Reynolds number has been measured, and it is later implied that this ratio is 0·6, whereas it is, of course, the ratio

$$\left[\frac{\varepsilon_H}{\nu}\right]_{\text{Sodium}} \text{to} \left[\frac{\varepsilon_M}{\nu}\right]_{\text{air}}$$

where ν = kinematic viscosity, that is 0·6; I think it would help if this were clarified in the paper. One would like to know a little more detail of the method used to deduce eddy diffusivity from the measured temperature and concentration profiles. Was any allowance made for the effect on the profiles of the varying convective component across the diameter of the pipe?

Whilst the velocity in the injection pipe was matched to that of the main flow, the turbulent structures would, of course, be quite different and would affect profiles in the early stages of diffusion, as would the property variation with temperature in the case of sodium; in the later stages there will be a considerable variation of eddy diffusivity across the breadth of the profiles. Does the ratio 0·6 refer to some average across the diameter of the pipe?

It is evident that, in view of the relative unimportance of eddy diffusion, such an average is adequate for the present purpose, but one wishes to avoid the figure being embodied in future liquid heat transfer theories if this is inappropriate.

On the general question of temperature variations across a complex heat transfer passage, such as a cluster, it is perhaps worth remarking that such variations, when judged in relation to the temperature difference from the heated surface to the local fluid, are worse with liquid metals than with higher Prandtl number fluids, such as gases. This was demonstrated in some work on a double annulus heat exchanger in connexion with the Dounreay fast reactor.* The reason is that eddy diffusion, which is mainly responsible for smoothing out transverse temperature variations in a cluster, or a non-uniform array of heated surfaces, is much less effective in relation to molecular diffusion of heat in liquid metals.

It appears to the writer that the transverse diffusion of heat under conditions appropriate to fast reactor cores is not particularly satisfactory. Taking a diffusivity of, say, 2 $cm^2\ s^{-1}$ (i.e. a Reynolds number of $\sim 5 \times 10^4$) the transverse distance to which a temperature disturbance penetrates across the flow is approximately $\sqrt{2t}$, where t is the diffusion time. If we take t as the time required for the coolant to pass through the core, i.e. $\sim$0·1 seconds this distance is $\sim$0·5 cm, which is small compared with the diameter of a cluster. It may be, of course, that such disturbances are expected to be small, but it is clear that considerable additional mixing would be required to materially

* Hall, W. B. and Jenkins, A. E., Heat transfer experiments with sodium potassium alloy, *J. Nucl. Energy Soc.* **1** (1955) 244–63.

affect the situation. A deliberately induced secondary flow would be more effective, and it would be interesting to know whether this has been considered.

Author's written reply

K. G. EICKHOFF—*Paper* 5B/5: The ratio $(\varepsilon_H/\varepsilon_M)$ quoted in the paper is the value for sodium. In evaluating this, however, it has been assumed that (ε_M/ν) for sodium is the same as that for air at the same Reynolds number. This assumption is considered to be reasonable.

The total thermal diffusivity for the central region of the flow has been measured, as discussed in the paper, by injecting a stream of sodium at 600°C into the main flow at 200°C. The decay of the temperature profile was analysed using the "moving point source" solution to obtain the total diffusivity. In this way a mean value for the central region of the flow is obtained, and from the total diffusivity the eddy contribution can be readily calculated.

It is true that close to the injector these measurements are not representative of the fully developed pipe flow, but the values quoted are taken from a region away from the injector.

Since the value of ε_M was obtained in the same way using an air facility, it is considered that the quoted value of $(\varepsilon_H/\varepsilon_M)$ adequately represents the conditions in the central region of the flow.

Professor Hall also refers to the small lateral penetration of a temperature disturbance in a subassembly. There are two temperature distribution effects which arise in a subassembly. Localized hot fluid streams, defined in the limit by three fuel pins and having a high temperature due, basically, to one or both of two sources, local flow and local heat generation variations, arise. In these, with the pin bundle geometry described in Paper 4B/1, the distance a temperature disturbance penetrates ($\sim$0·5 cm) is comparable to channel spacing and consequently diffusivity is effective in reducing hot channel effects. An overall temperature gradient can also result from an overall flux gradient. In this case the effect of diffusivity on the temperature distribution is marginal and appears as a flattening of the temperature profile at the edges of the subassembly.